

BGFM: BOLTZMANN-CALIBRATED FLOW MATCHING WITH UNIVERSAL NEURAL POTENTIALS FOR 3D MOLECULAR GENERATION

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ABSTRACT

Most de novo 3D molecular generators are trained to imitate molecular data. Their samples are often chemically valid but high in strain energy, requiring substantial downstream relaxation before use in simulation. Energy-based models target the equilibrium distribution directly but learn the energy from geometry alone and pay a heavy inference-time sampling cost. We propose **BGFM**, a Boltzmann-calibrated flow-matching framework for de novo 3D molecular generation. BGFM consists of four modules trained jointly: (i) an amortized flow proposal over coordinates, atom types, and charges; (ii) Boltzmann regularization that aligns the flow’s implied score and conditional density with an external universal neural potential through force-score and density-energy consistency losses; (iii) a calibrated scalar energy head distilled from the same neural potential; and (iv) a short learned-energy Langevin corrector that refines the flow proposal toward the energy landscape using 20–100 steps. We further introduce a joint density estimator that combines a coordinate-conditional FFJORD integral with the log-probability of the discrete categorical path, so that BGFM’s training objective is well-defined over the full molecular configuration (atom counts, types, charges, coordinates) rather than coordinates alone. On QM9 and GEOM-Drugs, BGFM matches the standard validity/stability of recent flow-matching baselines while preserving higher uniqueness and diversity. Under independent GFN2-xTB and MMFF relaxation oracles that are not used in training, BGFM produces lower-strain geometries than energy-based and flow baselines at matched compute, and improves the Pareto frontier of physical quality vs. diversity vs. sampling cost.

1 INTRODUCTION

Three-dimensional molecular generation is increasingly used as the front end of computational pipelines that assume physically realistic geometries: ligand discovery, electrolyte screening, catalyst design, and conformer-based property prediction. Two families of generators currently dominate. Flow-matching and diffusion models (Hoogetboom et al., 2022; Xu et al., 2023; Vignac et al., 2023; Stark et al., 2024; Song et al., 2024) are trained to imitate molecular data and amortize sampling over a small number of ODE steps; they produce diverse, chemically valid molecules but their geometries are not aligned with any explicit energy landscape and can require tens to hundreds of optimization steps under a quantum-chemistry oracle before they become useful. Energy-based models such as EBMol (Anonymous, 2025a) learn a persistent scalar energy from geometry alone and sample with Mirror-Langevin dynamics and parallel tempering; this yields physically aligned samples but at substantially higher inference cost and with a measurable diversity collapse at high sampling compute.

We propose BGFM, a framework that takes the favorable properties of each family by construction. BGFM keeps an amortized flow proposal as the primary generator, but couples it during training to an external universal neural potential (OMol25 (Levine et al., 2025), covering 83 elements and approximately 10^8 DFT-quality configurations) through two losses that align the flow’s implied score and conditional density with the potential’s force and energy landscape. To recover the inference-

time benefits of an explicit energy landscape without its sampling cost, BGFM additionally trains a calibrated scalar energy head distilled from the same potential, and uses it for a short learned-energy Langevin corrector that runs only 20–100 steps after the flow proposal. The result is a single trained model that inherits the diversity and amortized sampling speed of flow matching while matching or improving on the physical quality of energy-based samplers, at a small fraction of their inference compute.

A second methodological contribution is a joint density estimator that closes a gap in earlier flow-matching density-energy formulations (Klein et al., 2023; Grathwohl et al., 2019; Chen et al., 2018). Coordinate-only continuous-flow log-density is well-defined but does not account for the discrete structure (atom counts, types, charges) that a de novo generator also samples. BGFM combines a coordinate-conditional FFJORD integral with the log-probability of the discrete categorical path so that the training objective is a function of the full molecular configuration, not its continuous slice alone. This makes the Boltzmann regularization well-defined across compositions and provides a principled basis for the energy-head calibration on heterogeneous chemistry.

Contributions.

1. **Framework.** A single trained 3D molecular generator combining an amortized flow proposal, external neural-potential Boltzmann regularization, a calibrated scalar energy head, and a short learned-energy corrector. To our knowledge BGFM is the first de novo 3D flow generator that calibrates an amortized molecular generator against an external universal neural potential through both force-score consistency and density-energy Boltzmann regularization, with a learned energy corrector providing fast inference-time refinement.
2. **Joint density estimator.** A hybrid discrete-continuous log-density of the form $\log p_\theta(x) = \log p_\theta(r \mid c) + \log p_\theta(c)$ that combines a coordinate FFJORD integral with the CTMC path probability of the discrete components. This lifts BGFM’s density-energy loss from a coordinate-conditional alignment to a full molecular alignment.
3. **Calibrated energy head.** A scalar head $\hat{E}_\psi$ trained jointly with the flow against neural-potential energy and forces, supplying an explicit, calibrated energy landscape that BGFM uses for ranking and refinement.
4. **Evaluation framework.** A six-experiment protocol that separates standard generative quality, transferable physical quality under independent oracles, the quality-diversity-compute Pareto frontier, mechanistic Boltzmann alignment, ablations, and negative controls; with honest accounting of flow NFEs, energy-head NFEs, oracle calls, wall-clock time, and failure rates.

2 RELATED WORK

3D molecular generation with flows and diffusion. EDM (Hoogeboom et al., 2022), GeoLDM (Xu et al., 2023), MiDi (Vignac et al., 2023), GeoBFN (Song et al., 2024), JoMo (Le et al., 2024), SemlaFlow (Irwin et al., 2024), and FlowMol/FlowMol3 (Stark et al., 2024) train sample-set quality objectives (validity, FCD, distance distributions). Concurrent work, FlexiFlow (Anonymous, 2025b), extends flow matching to joint sampling of molecular graphs and conformer ensembles and is the current standard-generation state of the art on QM9 and GEOM-Drugs. None of these methods incorporates an explicit energy landscape or an external neural-potential supervision signal.

Energy-based molecular generation. EBMol (Anonymous, 2025a) trains a time-unconditional EBM on 3D molecular data and samples with Mirror-Langevin dynamics and parallel tempering. EBMol’s central claim is that an explicit scalar energy enables physically meaningful sample ranking, filtering, and steering. BGFM addresses the same problem through a complementary route: it keeps the amortized flow generator and calibrates a scalar energy head from an external universal neural potential, recovering the explicit-energy capability without the inference-time sampling cost of an EBM.

Boltzmann generators and energy-based flow matching. Boltzmann Generators (Noé et al., 2019), Equivariant Flow Matching (Klein et al., 2023), Implicit Transfer Operator Learning

(Schreiner et al., 2023), Scalable Boltzmann Generators (Wang et al., 2025), and the Adjoint Schrödinger Bridge Sampler (Liu et al., 2025) target Boltzmann densities for fixed classical potentials on single systems or for fixed-topology conformer generation. Iterated energy-based flow matching (Anonymous, 2025c) trains continuous normalizing flows from unnormalized energy densities using a simulation-free objective; it addresses fixed continuous Boltzmann sampling on low-dimensional benchmarks (Gaussian mixtures, double-well systems) and is theoretically related to BGFM but targets a different problem (known-energy continuous sampling, not de novo molecule generation under an external universal neural-potential teacher).

Universal neural potentials. SchNet (Schütt et al., 2018), NequIP (Batzner et al., 2022), MACE-MP-0 (Battat et al., 2023), and OMol25/eSEN (Levine et al., 2025) act as DFT replacements for molecular dynamics and relaxation. BGFM uses OMol25 not as a downstream relaxer but as a training-time teacher that supplies energies and forces for the Boltzmann regularization and the energy-head calibration.

3 METHOD

We denote by $x = (c, r)$ a molecular configuration with discrete part c (atom counts N , per-atom types, per-atom formal charges) and continuous part $r \in \mathbb{R}^{3N}$ (coordinates). An external universal neural potential supplies an energy $E_{\text{NP}}(r, c)$ and its gradient $F_{\text{NP}}(r, c) = -\nabla_r E_{\text{NP}}(r, c)$ for any admissible x . Temperature is parameterized by $\beta = 1/(kT)$.

The Boltzmann distribution we target is the conditional Boltzmann at the level of coordinates:

$$p_B(r \mid c, T) = \frac{1}{Z_c(T)} \exp[-\beta E_{\text{NP}}(r, c)]. \quad (1)$$

BGFM combines four modules: a flow proposal (§3.1), a Boltzmann regularization (§3.2), a calibrated energy head (§3.3), and a learned-energy corrector (§3.4). A joint density estimator (§3.5) provides a hybrid discrete-continuous log-probability used by the Boltzmann regularization.

3.1 MODULE 1 — FLOW PROPOSAL

We instantiate the flow proposal on a FlowMol3-style backbone (Stark et al., 2024): positions follow a Gaussian-prior linear interpolant, atom types and charges follow a CTMC simplex prior, and a GVP-Transformer parameterizes the time-dependent velocity field $v_\theta(x_t, t)$. Given prior samples $x_0 \sim p_0$, the flow maps x_0 to a candidate molecule $x = (c, r)$ in $K_{\text{flow}} \in [100, 500]$ ODE steps:

$$x_0 \sim p_0, \quad x = \Phi_\theta(x_0). \quad (2)$$

The standard flow-matching reconstruction loss $\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, x_1, t} \|v_\theta(x_t, t) - (x_1 - x_0)\|^2$ is retained alongside CTMC reconstruction terms for the discrete channels. The flow proposal preserves diversity and amortized sampling.

3.2 MODULE 2 — BOLTZMANN REGULARIZATION

The Boltzmann regularization couples Φ_θ to the neural potential through two consistency losses.

Force-score consistency. The Boltzmann density satisfies $\nabla_r \log p_B = F_{\text{NP}}/kT$. From the velocity field we recover the implied score in closed form for the linear interpolant:

$$s_\theta(r_t, t) = \frac{t v_\theta(r_t, t) - r_t}{(1 - t) \sigma^2}. \quad (3)$$

At late path times $t_{\text{eval}} \in \{0.70, 0.80, 0.90\}$, where r_t is close to the data manifold, BGFM enforces

$$\mathcal{L}_{\text{force}} = \mathbb{E}_{m, t_{\text{eval}}} [1 - \cos(s_\theta(r_t^m, t_{\text{eval}}), F_{\text{NP}}(r_1^m, c_m)/kT)]. \quad (4)$$

Density-energy consistency. For each parent molecule m we precompute $K = 5$ perturbations at displacement scales $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å and obtain their neural-potential energies. The Boltzmann constancy condition $\log p_\theta + \beta E_{\text{NP}} = -\log Z_{c_m}$ implies that for fixed c_m the quantity $y_m^{(k)} = \log p_\theta(r_m^{(k)} | c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m)$ is constant in k . The variance form removes the unknown additive constant $\log Z_{c_m}$:

$$\mathcal{L}_{\text{dens}} = \mathbb{E}_m \text{Var}_k \left[y_m^{(k)} \right]. \quad (5)$$

A composition-conditioned offset stabilizer $a_\phi(c)$, with input only the atom-type histogram and total charge, anchors the absolute scale:

$$\mathcal{L}_{\text{anchor}} = \mathbb{E}_{m,k} \left(\log p_\theta(r_m^{(k)} | c_m) + \beta E_{\text{NP}}(r_m^{(k)}, c_m) + a_\phi(c_m) \right)^2. \quad (6)$$

3.3 MODULE 3 — CALIBRATED ENERGY HEAD

A scalar energy head $\hat{E}_\psi(r, c)$ is attached to the flow backbone and trained jointly against the same neural-potential teacher:

$$\mathcal{L}_{\text{head}} = \mathbb{E} \left| \hat{E}_\psi(r, c) - E_{\text{NP}}(r, c) \right| + \lambda_F \mathbb{E} \left\| -\nabla_r \hat{E}_\psi(r, c) - F_{\text{NP}}(r, c) \right\|^2. \quad (7)$$

$\hat{E}_\psi$ is an explicit, calibrated energy landscape that BGFM uses (i) for inference-time refinement (§3.4), (ii) as a cross-model scorer for ranking generated samples (§4.7), and (iii) as the primary external counterweight to EBMol’s learned-but-uncalibrated energy.

3.4 MODULE 4 — LEARNED-ENERGY CORRECTOR

After the flow proposal returns a candidate (c, r) , BGFM applies a short Langevin corrector under $\hat{E}_\psi$:

$$r^{j+1} = r^j - \eta_j \nabla_r \hat{E}_\psi(r^j, c) + \sigma_j \xi^j, \quad \xi^j \sim \mathcal{N}(0, I), \quad j = 0, \dots, J-1, \quad (8)$$

with $J \in [20, 100]$, step size η_j following a small inverse schedule, and noise scale $\sigma_j = \sqrt{2\eta_j/\beta}$ so the chain targets $\exp(-\beta \hat{E}_\psi)$. Optional Metropolis correction can be enabled. The discrete components c are not modified; the corrector refines coordinates only. All applications of $\hat{E}_\psi$ are reported as energy-head NFEs, distinct from the flow ODE NFEs, in the experimental tables.

3.5 JOINT DENSITY ESTIMATOR

The Boltzmann regularization in Eqs. (5)–(6) requires $\log p_\theta$ on each perturbation. We estimate it as the sum of a coordinate-conditional log-density and a discrete log-probability:

$$\log p_\theta(x) = \log p_\theta(r | c) + \log p_\theta(c). \quad (9)$$

The coordinate term uses the FFFJORD instantaneous change-of-variables (Chen et al., 2018; Grathwohl et al., 2019):

$$\log p_\theta(r | c) = \log p_0(r_0) - \int_0^1 \nabla_r \cdot v_\theta(r_t, c, t) dt, \quad (10)$$

with the divergence estimated by Hutchinson’s trick (Hutchinson, 1990). The discrete term decomposes into contributions from atom counts N , per-atom types, and per-atom charges, evaluated on the CTMC bridge that generated c :

$$\log p_\theta(c) = \log p_\theta(N) + \sum_{i=1}^N \log p_\theta(a_i) + \sum_{i=1}^N \log p_\theta(q_i), \quad (11)$$

where each per-atom term is the bridge path probability under the CTMC parameterization (Stark et al., 2024). Using the full joint density makes $\mathcal{L}_{\text{dens}}$ and $\mathcal{L}_{\text{anchor}}$ well-defined across heterogeneous compositions, not just within fixed-composition clouds.

3.6 TOTAL OBJECTIVE

The training objective combines the flow-matching reconstruction, the Boltzmann regularization, and the energy-head calibration:

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_1 \mathcal{L}_{\text{force}} + \lambda_2 \mathcal{L}_{\text{dens}} + \lambda_3 \mathcal{L}_{\text{anchor}} + \lambda_4 \mathcal{L}_{\text{head}}. \quad (12)$$

The Langevin corrector (§3.4) is an inference-time procedure and does not contribute to training. We ramp λ_1 through λ_4 linearly from zero over the first two percent of training to protect the flow reconstruction during the early phase. Default values, derived from a dimensional analysis at room temperature ($kT = 0.025$ eV), are $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$, $\lambda_3 = 10^{-12}$, and $\lambda_4 = 1.0$ for the energy-magnitude term with $\lambda_F = 0.1$ for the force-gradient term.

4 EXPERIMENTS

Experimental questions. Our experiments are designed to answer five questions. *Q1.* Does BGFM preserve the standard chemical validity and diversity expected from de novo 3D molecular generators? *Q2.* Does BGFM improve physical quality under *independent* evaluators that are not used in training? *Q3.* How does BGFM compare with energy-based and flow baselines on the joint quality–diversity–compute frontier? *Q4.* Are the gains caused by correct neural-potential signal, rather than by additional regularization or evaluation overlap? *Q5.* Does each module of BGFM contribute, and what does each module add?

Evaluation philosophy. Because BGFM is trained with neural-potential energies and forces, neural-potential-based correlation is reported as a mechanism diagnostic rather than as a primary external benchmark. The headline physical-quality results use independent evaluators (GFN2-xTB, MMFF94, and a DFT subset) that do not appear in any training loss. For every sampling-cost number we report flow NFEs, energy-head NFEs, neural-potential oracle calls during sampling (zero by construction for BGFM), wall-clock time, and failure rate.

Benchmarks. We evaluate on three regimes: the standard QM9 benchmark, the GEOM-Drugs benchmark (Axelrod & Gómez-Bombarelli, 2022) under the revised OpenBabel/RDKit protocol, and held-out OMol25 broad-chemistry slices stressing charge, atom count, element composition, and metal-containing systems.

Baselines. Standard generation baselines: EDM (Hoogeboom et al., 2022), GeoLDM (Xu et al., 2023), GeoBFN (Song et al., 2024), SemlaFlow (Irwin et al., 2024), FlowMol3 (Stark et al., 2024), FlexiFlow (Anonymous, 2025b), and EBMol (Anonymous, 2025a). We additionally include controlled BGFM ablations on the same backbone: the flow-matching baseline (no Boltzmann regularization, no head, no corrector); BGFM without the energy head and corrector; BGFM without the density-energy loss; BGFM without the force-score loss; BGFM without the corrector; and full BGFM.

4.1 STANDARD UNCONDITIONAL GENERATION

We generate 10,000 molecules per model, following the standard EDM-style protocol for QM9 and the revised OpenBabel/RDKit protocol for GEOM-Drugs. Table 1 summarizes the metrics; the headline observation is the joint behavior of validity and diversity rather than peak validity alone.

4.2 INDEPENDENT PHYSICAL QUALITY

For each generated molecule we run GFN2-xTB relaxation (Bannwarth et al., 2019) and record the energy decrease ΔE_{relax} , the initial force norm, the number of optimization steps, the convergence failure rate, and the RMSD after relaxation. On organic subsets we additionally report MMFF94 (Halgren, 1996) statistics. For 200–500 randomly chosen valid molecules per model, we perform DFT single-point calculations or short DFT relaxations. None of these evaluators participates in the BGFM training loss. We report both a valid-and-connected protocol matching prior structural-energy tables and an all-samples protocol that counts invalid, disconnected, and failed molecules.

Table 1: Standard unconditional generation metrics on QM9 and GEOM-Drugs (revised OpenBabel/RDKit protocol). NFE refers to flow ODE steps; for BGFM the corrector contribution is reported separately. Diversity is the Vendi score (Friedman & Dieng, 2023).

Model	Bench.	NFE	Atom $\uparrow$	Mol $\uparrow$	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	Div $\uparrow$
EDM	QM9	1000	—	—	—	—	—	—
SemlaFlow	QM9	100	—	—	—	—	—	—
FlowMol3	QM9	100	—	—	—	—	—	—
FlexiFlow	QM9	100	—	—	—	—	—	—
EBMol	QM9	1810	—	—	—	—	—	—
BGFM (ours)	QM9	100	—	—	—	—	—	—
EDM	GEOM	1000	—	—	—	—	—	—
SemlaFlow	GEOM	100	—	—	—	—	—	—
FlowMol3	GEOM	100	—	—	—	—	—	—
FlexiFlow	GEOM	100	—	—	—	—	—	—
EBMol	GEOM	7240	—	—	—	—	—	—
BGFM (ours)	GEOM	100	—	—	—	—	—	—

Table 2: Independent physical quality on GEOM-Drugs valid-and-connected samples. xTB and MMFF are not used in training. Lower is better for ΔE , force norm, steps, and failure rate.

Model	V-c $\uparrow$	med $\Delta E_{\text{xTB}} \downarrow$	mean $\Delta E_{\text{xTB}} \downarrow$	p90 $\downarrow$	$\ F\ \downarrow$	Steps $\downarrow$	Fail $\downarrow$	Div $\uparrow$
FlowMol3	—	—	—	—	—	—	—	—
SemlaFlow	—	—	—	—	—	—	—	—
FlexiFlow	—	—	—	—	—	—	—	—
EBMol-low	—	—	—	—	—	—	—	—
EBMol-high	—	—	—	—	—	—	—	—
BGFM w/o corrector	—	—	—	—	—	—	—	—
BGFM w/o density-energy	—	—	—	—	—	—	—	—
BGFM w/o force-score	—	—	—	—	—	—	—	—
BGFM (full)	—	—	—	—	—	—	—	—

4.3 QUALITY-DIVERSITY-COMPUTE FRONTIER

Energy-based samplers can improve sample quality by increasing Langevin or parallel-tempering compute, but at a cost in diversity. We compare methods at multiple compute budgets. For EBMol we evaluate the low-, medium-, and high-NFE configurations. For BGFM we report several settings of the corrector horizon J and step size schedule alongside the unrefined flow proposal. We plot xTB median ΔE_{relax} against wall-clock time, with Vendi diversity indicated by color and the valid-low-strain count indicated by point size.

We additionally summarize the frontier with a single throughput metric:

$$\text{VULS}_\tau = \frac{\#\{\text{valid, unique, low-strain samples with } \Delta E_{\text{xTB}} < \tau\}}{\text{GPU-hour}}, \quad (13)$$

for $\tau \in \{2, 5, 10\}$ kcal/mol. VULS_τ measures how many chemically valid, physically low-strain, non-duplicate molecules a method delivers per unit compute.

4.4 MECHANISM DIAGNOSTIC: BOLTZMANN ALIGNMENT

For held-out parent molecules, we construct perturbation clouds using families not identical to the training shards: unseen Gaussian noise levels, torsion rotations, bond stretches, angle bends, RDKit conformer perturbations, and short xTB or neural-potential Langevin perturbations. For each perturbation we evaluate the joint density $\log p_\theta(x)$ from equation 9 and compare to $-E_{\text{NP}}/kT$. We report within-parent centered Pearson correlation, Spearman correlation, slope, and slope calibration error. This experiment verifies the mechanism that BGFM trains for; it is not the headline benchmark, because it shares its evaluator with the training loss. Cross-oracle transfer to xTB and a DFT subset is also reported.

Table 3: Module-level ablations. Each row removes one component of BGFM and reports the standard validity, the independent xTB ΔE , the OMol25 mechanism correlation, the Vendi diversity, and the total sampling wall-clock relative to the full method.

Configuration	Validity $\uparrow$	xTB ΔE $\downarrow$	Mech. corr. $\uparrow$	Div $\uparrow$	Time
FM backbone	—	—	—	—	—
+ force-score	—	—	—	—	—
+ density-energy	—	—	—	—	—
+ energy head	—	—	—	—	—
+ corrector (full BGFM)	—	—	—	—	—

4.5 ABLATIONS

Table 3 reports the contribution of each module of BGFM at matched optimizer budget.

4.6 NEGATIVE CONTROLS

To rule out training-evaluation overlap as the source of the gains, we re-train BGFM on shards in which the per-perturbation energies have been permuted within each parent molecule (the multiset of energies a parent receives is unchanged but the assignment to geometries is randomized). We additionally train on shards in which the per-atom forces have been permuted across atoms, in which the temperature is mislabeled, and in which $\mathcal{L}_{\text{dens}}$ is replaced by a random scalar regularizer with the same weight schedule. The headline causal claim is that the lift on the independent xTB physical-quality metric appears only for real-label training, even though the shuffled-label runs may still optimize the neural-potential-aligned correlation metric.

4.7 ENERGY-BASED RANKING AND FILTERING

A pooled set of samples is constructed from EDM, GeoLDM, FlowMol3, EBMol, the FM baseline, BGFM, deliberately strained perturbations, and relaxed reference structures. We score each molecule with the EBMol learned energy, the neural-potential energy, the BGFM density score

$$S_{\text{BGFM}}^{\text{dens}}(x) = \log p_{\theta}(r \mid c) - \beta E_{\text{NP}}(r, c) - a_{\phi}(c), \quad (14)$$

and the calibrated BGFM energy head $\hat{E}_{\psi}(r, c)$. Ground-truth strain is GFN2-xTB relaxation ΔE , xTB force norms, and DFT subset energy decrease. We report Pearson, Spearman, AUROC for the top decile of high-strain molecules, and the median relaxation energy after retaining the top-scored 10%, 20%, and 50% of samples.

5 CONCLUSION

BGFM uses an external universal neural potential as a training-time teacher for a de novo 3D molecular flow generator, combining an amortized flow proposal, a Boltzmann regularization that aligns the flow’s implied score and joint density with the potential, a calibrated scalar energy head distilled from the same potential, and a short learned-energy corrector that refines the flow proposal at inference time. The method preserves the diversity and amortized sampling speed of flow matching while recovering the explicit energy landscape that motivates energy-based generators, without inheriting their inference-time sampling cost. A hybrid discrete-continuous joint density estimator makes the regularization well-defined across heterogeneous chemistry.

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A DERIVATIONS

A.1 SCORE FROM FLOW-MATCHING VELOCITY

For the linear interpolant $r_t = (1 - t)r_0 + tr_1$ with $r_0 \sim \mathcal{N}(0, \sigma^2 I)$ and $r_1 \sim p_{\text{data}}$, the conditional distribution of r_0 given r_t is Gaussian with mean $r_t - tv^*$ and variance $(1 - t)^2 \sigma^2$. Marginalization and the identity $v^*(r_t, t) = \mathbb{E}[r_1 - r_0 \mid r_t]$ yield $s(r_t, t) = (tv^* - r_t)/[(1 - t)\sigma^2]$. Substituting v_θ for v^* gives equation 3.

A.2 VARIANCE LOSS AND THE PARTITION FUNCTION

For any r -independent shift c_m , $\text{Var}_k(y_m^{(k)} + c_m) = \text{Var}_k(y_m^{(k)})$ by translation invariance. Setting $c_m = -\log Z_{c_m}$ shows that the Boltzmann constancy condition is exactly the minimizer of equation 5, with the unknown partition function absorbed.

A.3 JOINT DISCRETE-CONTINUOUS LOG-DENSITY

The flow on (r, c) factors as a continuous coordinate flow conditioned on the CTMC discrete state plus a categorical bridge for the discrete state itself. By the chain rule for densities, $\log p_\theta(x) = \log p_\theta(r \mid c) + \log p_\theta(c)$. The first term is the FFIJORD integral equation 10 along the coordinate flow with c frozen at the endpoint label. The second term is the per-token CTMC path probability under the discrete schedule of Stark et al. (2024), summed over N , the per-atom type tokens, and the per-atom charge tokens, as in equation 11.

B HYPERPARAMETERS

Architecture. GVP-Transformer from FlowMol3: $n_{\text{hidden_scalars}} = 256$, $n_{\text{hidden_edge}} = 128$, $n_{\text{vec_channels}} = 32$, $n_{\text{molecule_updates}} = 6$, $n_{\text{recycles}} = 1$. The scalar energy head is a small invariant MLP on the final scalar features ($\sim 100k$ parameters).

Optimization. AdamW with $\text{lr} = 2 \times 10^{-4}$, weight decay 10^{-12} , warm-up over 2% of training, bf16-mixed precision. Batch size 16 with gradient accumulation 2.

BGFM hyperparameters. $\lambda_1 = 0.05$, $\lambda_2 = 10^{-4}$, $\lambda_3 = 10^{-12}$, $\lambda_4 = 1.0$, $\lambda_F = 0.1$. Temperature $kT = 0.025$ eV. Probe times $t_{\text{eval}} = \{0.7, 0.8, 0.9\}$. FFIJORD: $K_{\text{ODE}} = 4$, $H = 4$ Hutchinson samples. Energy-loss frequency: every fourth step. Perturbation $\sigma_p \in \{0.03, 0.06, 0.10, 0.20, 0.40\}$ Å, $K = 5$. Corrector: $J \in \{20, 50, 100\}$ at inference; step size η_j scaled by an inverse schedule from 10^{-3} to 10^{-4} ; $\sigma_j = \sqrt{2\eta_j/\beta}$.

C COMPUTE ACCOUNTING

For every BGFM sampling configuration we report (i) the number of flow ODE steps K_{flow} , (ii) the number of corrector steps J , (iii) the total number of energy-head forward passes $K_{\text{flow}} + J$ if $\hat{E}_\psi$ is also used to guide the flow (zero otherwise plus J for the corrector), (iv) the number of external neural-potential or xTB oracle calls during sampling (zero for BGFM by construction; reported as positive only for ablations that explicitly query the oracle online), (v) wall-clock time per sample on the reference hardware, and (vi) the number of failed samples that are dropped or replaced. EBMol’s

reported NFE budgets {930, 1370, 1810} on QM9 and {1080, 1960, 3720, 7240} on GEOM-Drugs are reproduced with the same accounting conventions.